

CASE STUDY · PEOPLE ANALYTICS

Uncovering the Real Drivers of Attrition

A worked walkthrough of why "employees are leaving because of pay" is usually the wrong answer and the analysis that finds the right one. Pairs well with the HR Employee Attrition practice dataset.

The Scenario

HR reports attrition has crept up to 18% annually, above the 12% the company considers healthy. The working theory in leadership is "we're not paying competitively." Before approving a company-wide salary review, they want the data to actually confirm it.

Step 1 — Resist the Single-Cause Story

Pay is the easiest explanation to reach for because it's the easiest one to picture fixing. That doesn't make it correct. The right first move is to pull every dimension available department, tenure, manager, performance rating, commute distance, overtime hours, promotion history and look at attrition rate split across each one, not just accept the headline theory.

Step 2 — Let the Data Rule Out, Not Just Confirm

A useful habit: actively try to disprove the leading theory before accepting it. Compare attrition rate for below-market-pay employees against at-or-above-market employees.

Segment	Attrition Rate
Below-market pay	19%
At/above-market pay	17%

A gap this small is a real finding in itself: pay is not the dominant driver. If leadership had approved a blanket raise on the original theory, it would have been an expensive fix for the wrong problem.

Step 3 — Find Where the Real Gap Actually Is

Continue the same comparison across every other dimension. In an analysis like this, the pattern that usually does separate cleanly is tenure under a specific manager, or overtime load:

Segment	Attrition Rate
Employees with regular overtime (>5 hrs/week)	31%
Employees with minimal overtime	11%

That's a far larger, more actionable gap than pay ever showed and it points toward a workload/staffing fix rather than a compensation fix.

Step 4 — Check for a Confound Before Declaring Victory

One more check matter here: is overtime itself the cause or is it a proxy for something else like understaffed teams or a specific manager with unusually high workload expectations? Cross overtime against department and manager before finalizing the recommendation. This is the same discipline as checking for Simpson's Paradox: a pattern that looks clean in aggregate can still be driven by one or two outlier teams.

Step 5 — Recommend the Fix That Matches the Cause

The final recommendation should map directly to what the data actually showed, not to the original assumption:

- Investigate staffing levels in the highest-overtime teams first, a hiring or redistribution fix, not a pay fix
- If specific managers show disproportionate overtime across multiple teams, that's a coaching/workload-planning conversation
- Keep compensation as a secondary lever, not the headline fix, since the data didn't support it as the primary driver

What This Case Study Teaches

The most valuable thing an analyst does in a case like this isn't confirming the obvious theory, it's being willing to rule it out and follow the data somewhere less convenient to leadership's first instinct.

This is a practice walkthrough for learning analytical structure — not a record of an actual client engagement. Terence Rebello · The Analyst's Toolbox